R_Assignment 1

Snehal

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Question 1: (60 points)

1.1

Generate a vector x from 5 to 35 with increment 2, and calculate its length.

```
x <- seq(from = 5, to = 35, by = 2)
x
```

```
##  [1]  5  7  9 11 13 15 17 19 21 23 25 27 29 31 33 35
```

```
length(x)
```

```
## [1] 16
```

1.2

Use the vector x in 1.1 to generate a 4-by-4 matrix A which filled by rows.

```
A <- matrix(x, nrow = 4, ncol = 4, byrow = TRUE)
A
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    5    7    9   11
## [2,]   13   15   17   19
## [3,]   21   23   25   27
## [4,]   29   31   33   35
```

1.3

Calculate the eigenvalues of the matrix A in 1.2.

```
eigen(A)$values
```

```
## [1]  8.381780e+01+0.000000e+00i -3.817805e+00+0.000000e+00i
## [3] -3.248838e-15+3.053487e-15i -3.248838e-15-3.053487e-15i
```

1.4

Change the 4 elements in first two rows and first two columns of the matrix A to 7. i.e. Let a11, a12, a21, a22 equal to 7.

```
A[1:2, 1:2] <- 7
A
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    7    7    9   11
## [2,]    7    7   17   19
## [3,]   21   23   25   27
## [4,]   29   31   33   35
```

1.5

Calculate the determinant of A in 1.4.

```
det(A)
```

```
## [1] 256
```

1.6

Calculate the inverse of A in 1.4.

```
A_inv <- solve(A)
A_inv
```

```
##        [,1]          [,2]    [,3]    [,4]
## [1,]  0.500 -6.278503e-16 -1.5000  1.0000
## [2,] -0.375 -1.250000e-01  1.3750 -0.8750
## [3,] -0.750  2.500000e-01 -0.4375  0.4375
## [4,]  0.625 -1.250000e-01  0.4375 -0.4375
```

1.7

Create a vector b by assigning the first row of A in 1.4 to b.

```
b <- A[1, ]
b
```

```
## [1]  7  7  9 11
```

1.8

Find y by solving linear equation A y = b with the A in 1.4 and b in 1.7. (Hint. y can be found by y = A-1 b where A-1 is the inverse of A.)

```r
y <- A_inv %*% b
y
```

```
##         [,1]
## [1,]  1.000
## [2,] -0.750
## [3,] -2.625
## [4,]  2.625
```

1.9

For each element of y in 1.8 find the minimum between its value and pi/2. Store all results into a single vector. Print the value of resulting vector.

```r
vmin <- pmin(as.vector(y), pi / 2)
vmin
```

```
## [1]  1.000000 -0.750000 -2.625000  1.570796
```

1.10

Read the documentation for function diag and use the function to generate the following 10-by-10 square matrix.

```r
diag(1:10)
```

```
##       [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
##  [1,]    1    0    0    0    0    0    0    0    0     0
##  [2,]    0    2    0    0    0    0    0    0    0     0
##  [3,]    0    0    3    0    0    0    0    0    0     0
##  [4,]    0    0    0    4    0    0    0    0    0     0
##  [5,]    0    0    0    0    5    0    0    0    0     0
##  [6,]    0    0    0    0    0    6    0    0    0     0
##  [7,]    0    0    0    0    0    0    7    0    0     0
##  [8,]    0    0    0    0    0    0    0    8    0     0
##  [9,]    0    0    0    0    0    0    0    0    9     0
## [10,]    0    0    0    0    0    0    0    0    0    10
```

Question 2

Consider a Fibonacci sequence Sn = Sn-1 + Sn-2 with the initial value S0 = 0 and the value at stage 1 as S1 = 1. Please determine the values of S3 and S50.

```r
fib <- function(n) {
  if (n == 0) return(0)
  prev <- 0
  curr <- 1
  if (n >= 2) {
```

```
    for (i in 2:n) {
      nxt  <- prev + curr
      prev <- curr
      curr <- nxt
    }
  }
  curr
}

fib(3)
```

```
## [1] 2
```

```
format(fib(50), scientific = FALSE, big.mark = ",")
```

```
## [1] "12,586,269,025"
```

Question 3

Find all the integers between 1 and 100 which are divisible by both 3 and 5. Store the results into a vector. Print each element of the resulting vector.

```
nums <- 1:100
result <- nums[nums %% 3 == 0 & nums %% 5 == 0]
for (v in result) {
  print(v)
}
```

```
## [1] 15
## [1] 30
## [1] 45
## [1] 60
## [1] 75
## [1] 90
```

Question 4

Create a function with input parameter n and returns a vector. The output vector contains all integers between 1 and n which are divisible by 3 and 5. Please test the function with two cases n = 100 and n = 200.

```
div_3_5 <- function(n) {
  v <- 1:n
  v[v %% 3 == 0 & v %% 5 == 0]
}
div_3_5(100)
```

```
## [1] 15 30 45 60 75 90
```

```r
div_3_5(200)
```

```
##  [1]  15  30  45  60  75  90 105 120 135 150 165 180 195
```

Question 5

Create a function with parameters a and b. In the function body, it tries to find the smallest positive number that is divisible by both a and b. Please test your function with following two cases (a = 3, b = 5) and (a = 6, b = 10).

```r
smallest_common_multiple <- function(a, b) {
  candidate <- max(a, b)
  while (candidate %% a != 0 || candidate %% b != 0) {
    candidate <- candidate + 1
  }
  candidate
}

smallest_common_multiple(3, 5)
```

```
## [1] 15
```

```r
smallest_common_multiple(6, 10)
```

```
## [1] 30
```